Artificial Intelligence for Video Games

Tidal Walking

Project due Dec. 9, 2025

Goal

The goal of this project is to create an agent that can traverse a hill-shaped landscape while avoiding being submerged by water.

Setup

The landscape must be generated using perlin noise to create a landscape with height between 0 and 10 meters. The size of the landscape is 50 by 100 meters. You can use our codebase for this.

In the landscape there is a water body whose level is changing in time. The level of the water will change between 0.5 and 7 meters at a constant speed of 0.5 meters per second. Once the maximum is reached, the level will decrease to the minimum and then raise again in an infinite loop

The agent must traverse the landscape from one corner to the opposite one staying out of the water at all times.

The agent is moving at a speed of 1 meter per second when staying at the same level or going down a slope. The movement is reduced to 0.5 meters per second when going up a slope (moving into higher ground).

Constraints

You must make sure that both the start and the goal corner are above 7.5 meters. They must always be out of the water.

The agent cannot stop. I.e., the agent MUST always move somewhere.